

Probability Theory and Statistics

Exercise 9 solution

11.03. – 11.07.

Covariance and correlation of continuous random variables, linear regression

1. By linearity of expectation,

$$\mathbb{E}(5X - 6Y) = 5\mathbb{E}(X) - 6\mathbb{E}(Y) = 5 \cdot 4 - 6 \cdot 0 = 20.$$

For independent random variables, the expectation of the product equals the product of expectations, hence

$$\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y) = 4 \cdot 0 = 0.$$

Variance is translation-invariant, so $\mathbb{D}^2(5X - 6Y + 8) = \mathbb{D}^2(5X - 6Y)$. For independent variables, the variance of a sum/difference is the sum of variances. Since X and Y are independent, so are $5X$ and $6Y$, thus

$$\mathbb{D}^2(5X - 6Y) = \mathbb{D}^2(5X) + \mathbb{D}^2(6Y) = 25\mathbb{D}^2(X) + 36\mathbb{D}^2(Y) = 25 \cdot 1 + 36 \cdot 2 = 97.$$

Covariance is also translation-invariant, hence

$$\text{cov}(5X + 2Y + 2, X + 6Y - 3) = \text{cov}(5X + 2Y, X + 6Y).$$

By bilinearity,

$$\text{cov}(5X + 2Y, X + 6Y) = 5\text{cov}(X, X) + 30\text{cov}(X, Y) + 2\text{cov}(Y, X) + 12\text{cov}(Y, Y).$$

Independence gives $\text{cov}(X, Y) = \text{cov}(Y, X) = 0$, while $\text{cov}(X, X) = \mathbb{D}^2(X) = 1$ and $\text{cov}(Y, Y) = \mathbb{D}^2(Y) = 2$. Therefore

$$\text{cov}(5X + 2Y + 2, X + 6Y - 3) = 5 \cdot 1 + 12 \cdot 2 = 29.$$

2. For $X, Y \sim U(0, 1)$ independent, we have $\mathbb{E}(X) = \mathbb{E}(Y) = \frac{1}{2}$ and $\mathbb{D}^2(X) = \mathbb{D}^2(Y) = \frac{1}{12}$.

a) Using $Z_1 = XY$ and independence,

$$\mathbb{E}(Z_1) = \mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}, \quad \mathbb{E}(Z_2) = \mathbb{E}(X + Y) = \mathbb{E}(X) + \mathbb{E}(Y) = 1.$$

b) Since $\mathbb{D}^2(XY) = \mathbb{E}[X^2Y^2] - [\mathbb{E}(XY)]^2$ and independence implies $\mathbb{E}[X^2Y^2] = \mathbb{E}(X^2)\mathbb{E}(Y^2)$,

$$\mathbb{E}(X^2) = \mathbb{E}(Y^2) = \mathbb{D}^2(X) + \mathbb{E}(X)^2 = \frac{1}{12} + \frac{1}{4} = \frac{1}{3}.$$

Hence

$$\mathbb{D}^2(XY) = \frac{1}{3} \cdot \frac{1}{3} - \left(\frac{1}{4}\right)^2 = \frac{7}{144} \neq \mathbb{D}^2(X)\mathbb{D}^2(Y) = \frac{1}{12} \cdot \frac{1}{12} = \frac{1}{144}.$$

So $\mathbb{D}^2(XY) \neq \mathbb{D}^2(X)\mathbb{D}^2(Y)$.

c) Using $\mathbb{D}^2(Z_1 + 2Z_2) = \mathbb{D}^2(Z_1) + \mathbb{D}^2(2Z_2) + 2\text{cov}(Z_1, 2Z_2)$,

$$\mathbb{D}^2(Z_1) = \frac{7}{144}, \quad \mathbb{D}^2(2Z_2) = 4\mathbb{D}^2(Z_2) = 4\mathbb{D}^2(X + Y) = 4\left(\frac{1}{12} + \frac{1}{12}\right) = \frac{2}{3}.$$

Moreover,

$$\text{cov}(Z_1, Z_2) = \mathbb{E}[XY(X + Y)] - \mathbb{E}(Z_1)\mathbb{E}(Z_2) = \mathbb{E}(X^2)\mathbb{E}(Y) + \mathbb{E}(X)\mathbb{E}(Y^2) - \frac{1}{4} \cdot 1 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12},$$

so $\text{cov}(Z_1, 2Z_2) = 2 \cdot \frac{1}{12} = \frac{1}{6}$. Therefore

$$\mathbb{D}^2(Z_1 + 2Z_2) = \frac{7}{144} + \frac{2}{3} + 2 \cdot \frac{1}{6} = \frac{151}{144} \approx 1.049.$$

3.

$$\text{cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y).$$

From the previous exercise we know $\mathbb{E}(X) = \mathbb{E}(Y) = \frac{13}{20}$. Compute

$$\mathbb{E}(XY) = \int_0^1 \int_0^1 xy \cdot 2(x^3 + y^3) dx dy = \int_0^1 \left[\frac{2}{5}y + \frac{1}{1}y^4 \right] dy = \left[\frac{1}{5}y^2 + \frac{1}{5}y^5 \right]_0^1 = \frac{2}{5}.$$

Hence

$$\text{cov}(X, Y) = \frac{2}{5} - \left(\frac{13}{20} \right)^2 = -\frac{9}{400} \quad (\approx -0.0225).$$

4. Since $Y = 3X + 8$ and $Z = 5 - 2X$ are linear functions of the same $X \sim N(m, \sigma^2)$,

$$\text{cov}(Y, Z) = \text{cov}(3X + 8, 5 - 2X) = \text{cov}(3X, -2X) = -6\mathbb{D}^2(X) = -6\sigma^2,$$

$$\mathbb{D}^2(Y) = \mathbb{D}^2(3X + 8) = 9\mathbb{D}^2(X) = 9\sigma^2, \quad \mathbb{D}^2(Z) = \mathbb{D}^2(5 - 2X) = 4\mathbb{D}^2(X) = 4\sigma^2.$$

Therefore

$$\text{corr}(Y, Z) = \frac{\text{cov}(Y, Z)}{\mathbb{D}(Y)\mathbb{D}(Z)} = \frac{-6\sigma^2}{(3\sigma)(2\sigma)} = -1.$$

5. Let $X \sim U(0, 2\pi)$, $Y = \cos X$, $Z = \sin X$.

$$\mathbb{E}(YZ) = \frac{1}{2\pi} \int_0^{2\pi} \sin x \cos x dx = \frac{1}{4\pi} \int_0^{2\pi} \sin(2x) dx = 0,$$

and similarly $\mathbb{E}(Y) = \mathbb{E}(Z) = 0$, hence $\text{cov}(Y, Z) = 0$. However, Y and Z are *not* independent (e.g., $Y^2 + Z^2 = 1$ a.s., so there is a deterministic relation).

6.

$$\text{cov}(U, V^2) = \mathbb{E}(UV^2) - \mathbb{E}(U)\mathbb{E}(V^2).$$

Using

$$\mathbb{E}(UV^2) = \int_0^1 \int_0^{u^2} u v^{3/2} dv du = \int_0^1 \frac{2}{5} u^6 du = \frac{2}{35},$$

$$\mathbb{E}(U) = \int_0^1 \int_0^{u^2} \frac{u}{\sqrt{v}} dv du = \int_0^1 2u^2 du = \frac{2}{3},$$

$$\mathbb{E}(V^2) = \int_0^1 \int_0^{u^2} v^{3/2} dv du = \int_0^1 \frac{2}{5} u^5 du = \frac{1}{15},$$

we obtain

$$\text{cov}(U, V^2) = \frac{2}{35} - \frac{2}{3} \cdot \frac{1}{15} = \frac{4}{315} \approx 0.01270.$$

7. **(a)** Since the variables are uncorrelated, any pair has covariance 0. By bilinearity and $\text{cov}(X_i, X_i) = \mathbb{D}^2(X_i) = 1$,

$$\text{cov}(X_1 + X_2, X_2 + X_3) = \mathbb{D}^2(X_2) = 1.$$

Moreover,

$$\mathbb{D}^2(X_1 + X_2) = \mathbb{D}^2(X_1) + \mathbb{D}^2(X_2) + 2\text{cov}(X_1, X_2) = 2, \quad \mathbb{D}^2(X_2 + X_3) = 2.$$

(b) Hence

$$\text{corr}(X_1 + X_2, X_2 + X_3) = \frac{1}{\sqrt{2}\sqrt{2}} = \frac{1}{2}.$$

(c) The regression line of $Y = X_2 + X_3$ on $Z = X_1 + X_2$ has

$$\beta = \frac{\text{cov}(Z, Y)}{\mathbb{D}^2(Z)} = \frac{1}{2}, \quad \alpha = \mathbb{E}(Y) - \beta \mathbb{E}(Z) = 0,$$

thus $y = \frac{1}{2}x$. **(d)** The mean squared error is

$$\mathbb{D}^2(Y - (\beta Z + \alpha)) = \mathbb{D}^2(Y) \left(1 - \text{corr}(Z, Y)^2\right) = 2 \left(1 - \left(\frac{1}{2}\right)^2\right) = \frac{3}{2}.$$

(Equivalently, expand variances and covariances directly to get $\frac{3}{2}$.)

8. We regress $X^2 + X + 1$ on X . In general,

$$\beta = \frac{\text{cov}(X^2 + X + 1, X)}{\mathbb{D}^2(X)} = \frac{\text{cov}(X^2, X) + \text{cov}(X, X)}{\mathbb{D}^2(X)} = \frac{\mathbb{E}(X^3) - \mathbb{E}(X)\mathbb{E}(X^2)}{\mathbb{D}^2(X)} + 1,$$

$$\alpha = \mathbb{E}(X^2) + \mathbb{E}(X) + 1 - \beta \mathbb{E}(X).$$

(a) $X \sim \text{Exp}(\lambda)$ (then substitute $\lambda = 5$): $\mathbb{E}(X) = 1/\lambda$, $\mathbb{D}^2(X) = 1/\lambda^2$, $\mathbb{E}(X^2) = 2/\lambda^2$, and by integration by parts $\mathbb{E}(X^3) = 6/\lambda^3$. Hence

$$\beta = \frac{6/\lambda^2 - (1/\lambda)(2/\lambda^2)}{1/\lambda^2} + 1 = \frac{4}{\lambda} + 1 \Rightarrow \beta = \frac{9}{5} \text{ for } \lambda = 5,$$

$$\alpha = \frac{2}{\lambda^2} + \frac{1}{\lambda} + 1 - \beta \frac{1}{\lambda} \Rightarrow \alpha = \frac{23}{25} \text{ for } \lambda = 5.$$

Thus $y = \frac{9}{5}x + \frac{23}{25}$. **(b)** $X \sim N(0, 1)$: $\mathbb{E}(X) = 0$, $\mathbb{D}^2(X) = 1$, $\mathbb{E}(X^2) = 1$, and by symmetry $\mathbb{E}(X^3) = 0$.

$$\beta = 1, \quad \alpha = 2, \quad \text{so } y = x + 2.$$

(c*) $X \sim \text{Bin}(10, 0.3)$: $\mathbb{E}(X) = 3$, $\mathbb{D}^2(X) = 2.1$, $\mathbb{E}(X^2) = 11.1$. Writing $X = \sum_{i=1}^n X_i$ with i.i.d. Bernoulli(p),

$$\mathbb{E}(X^3) = np + 3n(n-1)p^2 + n(n-1)(n-2)p^3 = 46.74.$$

Hence

$$\beta = \frac{46.74 - 3 \cdot 11.1}{2.1} + 1 = 7.4, \quad \alpha = 11.1 + 3 + 1 - 7.4 \cdot 3 = -7.1,$$

so $y = 7.4x - 7.1$.

9. We have $\mathbb{D}^2(X) = 0.0775 = \frac{31}{400}$ and, by symmetry, $\mathbb{D}^2(Y) = \frac{31}{400}$. **(a)** Using $\text{cov}(X, Y) = -\frac{9}{400}$,

$$\beta = \frac{\text{cov}(X, Y)}{\mathbb{D}^2(X)} = -\frac{9}{31}, \quad \alpha = \mathbb{E}(Y) - \beta \mathbb{E}(X) = \frac{26}{31}.$$

Thus the regression line of Y on X is $y = -\frac{9}{31}x + \frac{26}{31}$. The mean squared error is

$$\mathbb{D}^2(Y - (\beta X + \alpha)) = \mathbb{D}^2(Y) - \frac{\text{cov}(X, Y)^2}{\mathbb{D}^2(X)} = \frac{31}{400} - \frac{(-9/400)^2}{31/400} \approx 0.07097.$$

(b) By symmetry, the regression of X on Y has the same coefficients in swapped roles, i.e.

$$x = -\frac{9}{31}y + \frac{26}{31}, \quad \text{MSE} \approx 0.07097.$$

10. Given the regression line of Y on X is $y = x - 2$, we have

$$\frac{\text{cov}(X, Y)}{\mathbb{D}^2(X)} = 1, \quad \mathbb{E}(Y) - \mathbb{E}(X) = -2.$$

Given the regression of X on Y is $x = \frac{1}{4}y - \frac{11}{2}$,

$$\frac{\text{cov}(X, Y)}{\mathbb{D}^2(Y)} = \frac{1}{4}, \quad \mathbb{E}(X) - \frac{1}{4}\mathbb{E}(Y) = -\frac{11}{2}.$$

Solving

$$\mathbb{E}(Y) - \mathbb{E}(X) = -2, \quad \mathbb{E}(X) - \frac{1}{4}\mathbb{E}(Y) = -\frac{11}{2}$$

gives

$$\mathbb{E}(X) = -8, \quad \mathbb{E}(Y) = -10.$$

11. Two dice are rolled. Let X be the number of sixes and Y the number of even results. One computes

$$\mathbb{E}(XY) = \frac{1}{2}, \quad \mathbb{E}(X) = \frac{1}{3}, \quad \mathbb{D}^2(X) = \frac{5}{18}, \quad \mathbb{E}(Y) = 1, \quad \mathbb{D}^2(Y) = \frac{1}{2}.$$

Hence

$$\text{cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y) = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}.$$

Regression of Y on X :

$$\beta = \frac{\text{cov}(X, Y)}{\mathbb{D}^2(X)} = \frac{1/6}{5/18} = \frac{3}{5}, \quad \alpha = \mathbb{E}(Y) - \beta \mathbb{E}(X) = 1 - \frac{3}{5} \cdot \frac{1}{3} = \frac{4}{5},$$

thus $y = \frac{3}{5}x + \frac{4}{5}$. Regression of X on Y :

$$\beta = \frac{\text{cov}(X, Y)}{\mathbb{D}^2(Y)} = \frac{1}{3}, \quad \alpha = \mathbb{E}(X) - \beta \mathbb{E}(Y) = \frac{1}{3} - \frac{1}{3} \cdot 1 = 0,$$

thus $x = \frac{1}{3}y$.